From Prototype to Platform: Measuring the Commercial Viability Index of Digital Assets

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Abstract*— Prototypes of software can prove to be technically feasible, but not adequate evidence of development to become scalable and commercially viable digital platforms. The prevailing way to think about these aspects in existing work is to consider them as mutually distinct. In existing work these aspects are often discussed separately: software architecture, technical debt, API maturity, organizational readiness, platform governance, digital transformation capability, commercial performance. This paper introduces a measurement-based Commercial Viability Index (CVI) framework which combines six dimensions of readiness and regards the commercial performance as the external result. A series of evidence gates are applied sequentially for provenance, measurement quality, discriminant validity, collinearity, robustness and practical relevance, with the composite readiness score only interpreted after each of the evidence gates. It is used only as a methodological demonstration due to the fact that the indicators from the original SmartPLS European Customer Satisfaction Index do not apply to the constructs of the thesis. Construct correlations and inner variance inflation factors are high, suggesting that path-derived weights are not yet defensible, based on preliminary analysis. To standardize the score for the eligible five-point construct on a 0–100 scale and to exclude out-of-range and incomplete inputs, the scores are normalized in a transparent equal-weight reference implementation. In a transparent equal weight reference implementation, the scores are normalized on a 0–100 scale and out-of-range or incomplete inputs are rejected. There were 5 pre-defined Python unit tests which passed but exhibited arithmetic and failure-state behavior, not empirical validity. The contribution is an auditable framework that differentiates the design, implementation, testing and validation processes, thereby minimizing the risk of presenting unstable statistical relationships as index for commercial value.

Keywords*— Commercial Viability Index (CVI), digital assets, platform readiness, PLS-SEM, software architecture, technical debt, digital transformation and reproducibility.

I. INTRODUCTION

Digital assets are becoming more complex and are no longer just an application but have become more of a platform that can be used by more people, in the ecosystem and can be scaled to serve more use-cases [1, 2]. While a technically viable prototype is not necessarily commercially viable at platform scale. Even if the initial technical performance is good, architecture quality, technical debt, API capability, organization readiness, governance and transformation capability can limit scale [3]–[6]. The research problem is that these factors are typically measured in isolated technical, organizational and/or business areas. Software engineering focuses on the attributes of software, such as quality, maintainability, scalability, and debt, while information-systems and platform research focuses on the capability of the organization, governance, ecosystems and value created through information systems [1, 2, 5, 6]. This separation hinders the ability to make a clear determination whether a software prototype is ready in one clear evaluation to move forward to a viable digital platform.

This paper proposes a measurement first research artefact called Commercial Viability Index (CVI). It is not a question of creating a single nice score without any qualms. Rather, there needs to be evidence that each construct is traceable, that it is measured in a consistent way and that it is distinct and appropriate for aggregation, before the readiness score can be considered meaningful. Such a design directly addresses the preliminary empirical results that very high correlation and/or collinearity can cause model structural coefficients to be unstable even when the model appears to be a good fit [7, 8]. The paper has three limited contributions. First, it introduces a readiness architecture that is based on six dimensions and it leaves the Commercial Performance Indicator (CPI) as a result rather than an element of the readiness aggregate. Second, it defines sequential validation gates that will not allow unsupported weighting and interpretation. Third, it offers a reproducible equal-weight Python reference implementation, rejection rules and pre-defined tests to check it. Thus, the present work is not a validation study in terms of a complete validation of a market readiness instrument, but a framework and implementation study, including preliminary empirical evidence.

II. BACKGROUND AND RESEARCH GAP

A. Prototype-to-Platform Readiness

Research on digital transformation focuses on the change of the organization due to digital technologies and research on platform is about the co-evolution of architectures, governance and ecosystems [1, 2, 6]. The architecture of the software can impact its modifiability, performance, security and deployability [3]. Technical debt can be a means to speed up delivery in the short term, but come with a cost in the long term and decrease maintainability if not managed [4]. Maturity in APIs has an impact on their interoperability and stable offering to partners and complementary services. These technical characteristics are not independent or stand-alone, but instead are intersected by organizational readiness, governance and the capability of organizational transformation. While there are good sources in the literature to support the dimensions singly, the thesis review did not find a single source in the reviewed literature that

covers the dimensions of software architecture quality, software technical debt, API maturity, organizational readiness, platform governance, digital transformation capability, and a commercial outcome in a single readiness assessment. The difference lies both in the integrative and the measurement aspects: the question is which is the most effective means of combining a series of multidimensional evidence without losing the weak nature of the measurement or being misled by unstable relationships. C. When to use composite readiness scores However, composite indicators are highly useful as they help to summarize multidimensional information; they can be constructed in a transparent way, making choices regarding normalization, direction, eligibility, weighting, and sensitivity [9]. To obtain latent-variable modelling, the reliability is not enough. Before interpreting structural paths, the convergence of data, the discrimination of data and the collinearity of the data must be assessed [7], [8]. For this reason, the proposed CVI does not see validation as a step to follow scoring; instead, it is considered to be a gate.

III. RESEARCH DESIGN AND METHOD

This study used a pragmatic design of study in the form of the development of a framework and validation of the study with empirical data. The quantitative workflow involves descriptive screening and reliability analysis and then partial least squares structural equation modelling (PLS-SEM). The planned order of the tests is indicator assessment, internal-consistency reliability, convergent validity (average variance extracted (AVE)), discriminant validity (heterotrait-monotrait ratio (HTMT)), collinearity (variance inflation factors (VIF)) and finally the structural interpretation with bootstrapping [7, 8]. There are 5,000 bootstrapped resamples for the planned inferential procedure.

A. Data Source

Internal Data Source The only current benchmark is the SmartPLS European Customer Satisfaction Index (ECSI) sample project (250 observations) of the French mobile-phone industry and related to the PLS path-modelling example of Tenenhaus et al. [10]. This data set has not been specifically developed to track commercial readiness for digital-assets. In this regard, the thesis variables cannot be seen as direct measures of the constructs without an auditable item level mapping. This benchmark is presented here as an example for the analytical and implementation pipeline, but as a non-generalization to software organizations in Australia. Therefore, a provenance matrix is needed in the process of final empirical interpretation. The record should include the original indicator and text, response anchors, source construct, proposed CVI construct, any recoding/reversal, should include a theoretical rationale for mapping, with a verification status. Technical Debt Management (TDM) direction should be decided based on the semantic meaning of the items and coding, but not on the signs of the correlations.

B. Candidate Constructs

Role	Construct	Purpose in framework
Readiness	Software Architecture Quality (SAQ)	Architecture quality supporting scalable evolution
Readiness	Technical Debt Management (TDM)	Debt burden/management capability after direction audit
Readiness	The API Maturity (APIM)	interoperability and the reusable service capability of the API
Readiness	Organizational Readiness (OR)	ability to support platform transition.
Readiness	Platform Governance (PG)	Consulting with stakeholders and developing PG Rules, control, accountability and ecosystem coordination
Readiness	Digital Transformation Capability (DTC)	Capability to perform and maintain digital transformation
Outcome	Commercial Performance Indicator(CPI)	Commercial outcome outside of the readiness aggregate; excluded from the commercial outcome

TABLE I. CANDIDATE CVI CONSTRUCTS AND ROLES

IV. MEASUREMENT-FIRST CVI FRAMEWORK

A. Validation-Gated Architecture

The CVI is carried out as a series of steps: Logical design means that the stage logic and requirements are described and implemented; executable means that there are executable artefacts; test means that there are cases described in the stage and the acceptance requirements are satisfied, and validation means that there are appropriate empirical/practical evidence supporting the stage. This separation makes it difficult to have work presented as proof of what is done.

Gate	Question	Decision
1. Provenance	Do source items, coding and mappings have been verified?	If there is no resolution on the origin, discontinue the process.
2. Measurement quality	is reliability and convergent validity good?	Edit or remove poor constructions.
3. Distinctness	are there any supporting diagnostics and/or any HTMT that suggest discriminant validity?	Be careful not to just add evidence without considering.
4. Collinearity	Is the interpretation of the predictor VIFs stable?	Avoid using unstable paths to be used as weights
5. Robustness	Do the results of the comparators and sensitivities remain consistent?	If not, go for transparent baseline
6. Relevance	Does it have a practical purpose and is backed up with appropriate evidence?	Do not place any limits on claims until validated practically

TABLE II. SEQUENTIAL EVIDENCE GATES

B. Reference Scoring Logic

If it is an eligible construct that is measured on a validated 5-point scale, the reference normalizations is (1).

Normalized score = ((construct score - 1) / 4) × 100 (1)

Once directional alignment is done, the reference CVI for observation i is the average of the six readiness dimensions that are eligible for CVI. CPI is not included as it is considered to be an outcome, not a readiness measure. If a construct is missing or is not in the valid range, the implementation is not silent about the missing construct, but instead fails to perform the full calculation or returns the result of the calculation explicitly qualified. When no empirical defensible basis for differential weights exists, equal weighting is considered to be the baseline.

V. RESULTS AND IMPLEMENTATION

A. Preliminary Statistical Pattern

The earlier analysis reported construct means of SAQ = 3.897, APIM = 3.894, CPI = 3.851, DTC = 3.845, PG = 3.824, OR = 3.805, and TDM = 2.558. The magnitude of the correlation with the CPI was around 0.983–0.994, and the value of R^2 for the CPI was reported as 0.991. The values of inner VIF were around 50.9–153.4. The values are therefore only preliminary evidence as they could not be checked in the thesis revision by the underlying SmartPLS export files. The problem of measurement is not solved due to the high R^2. The extremely high correlations and VIF values suggest that there is much overlap among the predictors, and each path coefficient is not a reliable CVI weight. The framework does not, therefore, attempt to make these early structural coefficients a weighted commercial-readiness score.

The next priority in the empirical agenda is to create construct provenance and distinctness and then re-execute measurement and structural models with known model settings and outputs. B. Reproducible Reference Implementation The reference scoring logic has been created in Python code that explicitly validates input and provides for failure states. Five different unit tests were used: tests for boundaries of the scale; test for an example in the middle of the range; test for rejection of the unit with missing constructs; test for rejection of the unit outside of the scale; test for reproducing the previously reported sample-means demonstration. 5 tests passed. With the reported construct means and a provisional TDM reversal, the demonstration yields the values of 69.6125/100. This result is only for the arithmetic of the implementation, not the construct mapping, thresholds, prediction or commercialization of the benchmark observations.

Evidence item	Current status	Interpretation
Dataset structure	250 observations used for benchmark	Appropriate for pipeline demonstration but needs to have a provenance map added.
Measurement/structural diagnostics	Preliminary	If there is a high correlation and VIF, the should be re-evaluated.
Supported Equal-weight CVI logic	Implemented	Transparent reference comparator
The tests for Python	5/5 passed	the arithmetic and failure-state behaviour tests.
Practical/expert validation	Not completed	No market-readiness claim

TABLE III. CURRENT EVIDENCE STATUS

VI. DISCUSSION

Central to the methodology is the idea that an index that is to be used in the commercial world should not outpace the evidence it is based upon. There is scope for a model to have high explanatory fit, but have constructs that are too intercorrelated to be interpreted separately. Therefore, the preliminary correlations and VIFs in this study are considered as a warning instead of confirmation of dimensions.

This alters the function of PLS-SEM from a step to automatically generate weights to a diagnostic and validation step to see if weights can be justified. The measurement first design also enhances the traceability. There is a connection between each research question/ objective and a construct, evidence source, analytical procedure, artefact and decision rule. The CVI that is created is not just a formula, it's a controlled process in that it only emits the score when the required evidence for interpretation is in place. This is particularly important when secondary data are employed in methodological demonstration since it is not possible to replace the construct meaning with a plausible statistical pattern.

The framework is designed to be leveraged for structured readiness discussion for practitioners—before significant investments in platforms. The current version does not yet allow for using dimension profiles or explicit warnings for investment decisions. Later, with the data and constructs that fit their respective provenance and, if their data and constructs involve human subjects and/or organizations, meet the necessary ethics and access approvals, they will be able to assess the practical usefulness, sensitivity and case-level interpretation of the data and constructs.

VII. LIMITATIONS AND FUTURE WORK

Construct validity is the one major drawback. In the ECSI benchmark, the six suggested dimensions of digital asset readiness are not modelled but rather the customer satisfaction. If an item-level mapping and coding audit is not conducted, the benchmark is unable to verify direct measurement of the thesis constructs. A second limitation is that the degree of construct overlap and collinearity are reported which makes independent structural effects and path derived weighting unstable.

The last constraint is that the underlying data and SmartPLS exports to present the preliminary results were not made independent in the thesis revision. There is also no yet practical or expert validation. The next stage should be done in dependency order: Verify the origins of the questionnaires and codebook, and item mapping; Ensure the coding direction, repeated-profile independence of the questionnaires; Re-run the measurement and structural diagnostics with documented software settings; Ensure the observation level CVI scores; Ensure comparable and defensible weighting schemes; Ensure sensitivity, uncertainty and negative-case analysis; and finally, ensure practical validation (if ethical and access requirements allow). The decision-support dashboard can be added once the measurement and scoring core is established and is stable.

VIII. EVALUATION AND DECISION RULES

A. Research Traceability

A central design requirement is traceability from the research problem to the evidence used in a decision. The research objectives are thus associated with each readiness dimension, a construct definition, an indicator source, an analytical procedure, and an output. This prevents the CVI from being considered as a bunch of independent tests. It also gives an obvious traceability in case a construct is changed or deleted. When an indicator is not found to be traceable to a defensible source, or its meaning cannot be interpreted in terms of the proposed construct, then such an indicator should not be included as a part of the final score.

The same holds true for structural paths: when the discriminant validity and/or the collinearity diagnostics are not statistically satisfied, a high path coefficient, but not necessarily a substantive meaningful contribution, is not interpreted as such. The framework separates the evidence related to measurement from the evidence of outcomes. The following dimensions are considered to be candidate readiness dimensions: SAQ, TDM, APIM, OR, PG, DTC. CPI is set up as an outcome, which can be tied to readiness, but does not count towards the readiness index. This separation tends to deplete circularity, which is a good standard indicator of commercial performance: a high-performance measure directly leading to a commercial performance is being separated from an indicator that is supposed to explain or assess for readiness to achieve a commercial performance. The architecture is therefore able to enable future assessment using predictive or criterion-related assessment, and provide a conceptually interpretable readiness score.

B. Comparator and Sensitivity Strategy

The equal-weight CVI is used to compare the network, since it is not opaque and doesn't need to be read as importance weights. Alternative weighting is only considered after the quality of the measurements is acceptable and when there is no collinearity, and when the alternative provides a stable and meaningful improvement. The comparison should be reported of the changes in the observation rankings, dimension contributions, substantive conclusions under reasonable perturbations of the weights. In cases where a small variation in weights results in a large difference in readiness classifications, then a sensitivity warning should be used rather than as a decision threshold. This strategy is based on the general composite-indicator principle that the weights should be explicitly stated and be subject to testing [9]. For all sensitivity analysis, both construct and decision level sensitivity analyses should be performed.

The analysis at the construct level looks at how realistic weight changes impact the results, how alternative treatment of the TDM after confirming the direction of the TDM affects the results, and how any construct that does not pass a validation gate is excluded. At the decision level, the analysis will consider whether the observations are still relatively high or low in readiness and whether any proposed categories are sound if there is some variations in modelling. The present document does not specify validated threshold values, so the reference implementation won't say that, e.g., a score of 69.6 is "ready" or "not ready. It provides an arithmetic score, and also a description of the evidence status, which will be used for interpretation.

C1. Failure states, Warning Logic

There is a need for a useful readiness tool that will communicate when it is not to be trusted. Therefore, the proposed implementation considers the following states to be failure or warning states: missing constructs, out-of-range values, out-of-range directionality, out-of-range measurement diagnostics, and out-of-range provenance. If there are missing dimensions, then no unqualified full CVI will be generated. When the value exceeds the validated scale or the number of valid responses exceeds the scale length, it is not clipped silently, but is rejected. When there is no direction in TDM, the system should not reverse the score just because it seems convenient to do so because of a negative relationship. If discriminant validity or collinearity is still not good enough, the score could still be displayed as a clear reference-composite, but with the omission of the structural weights and causal language. This is a warning logic and changes the usage of the index, so it is part of the contribution. Basic dashboards can lead the user to a one-line focus. The CVI design, on the other hand, correlates the number with a dimension profile, provenance status, validation status and warnings. A future dashboard should, therefore, show the evidence state along with the score and enable a user to view the rationale behind an evidence state of a construct (included, excluded, reversed, or flagged). This type of behavior helps to make the limitations transparent to decision makers and facilitates auditability over a long period of time, instead of being lost in technical documentation.

D. Reproducibility Procedure

It's not enough to just have a final spreadsheet to be reproducible. The research record should be able to contain the source dataset, the data dictionary, an analysis-ready version, software versions, settings for the algorithms and the bootstrapping, exported diagnostics and the decisions made for the indicators, the scoring code, the unit tests and a change log. Transformations should be deterministic, and there should be no changes made to the master source. Working copies are also versioned, and can be used to determine which data and which code caused a reported result. The procedure is especially relevant in the current study since the initial statistical values need to be differentiated from subsequent reruns which occur after the mapping of the provenance and constructs have been confirmed. The outer loadings, internal-consistency estimates, AVE, HTMT (if relevant), cross-loading or Fornell-Larcker evidence, inner VIF, path coefficients, bootstrap confidence intervals and t statistics, R^2, effect sizes and predictive diagnostics (if applicable) should be included in the final evidence package for the PLS-SEM stage. These outputs can all be reported together, so that a reviewer can determine if the measurement model supports the structural interpretation. An R^2 value is not considered to be a good enough proof. Additionally, the analysis record should include information about the indicators which were included and excluded in the analysis and its respective rationale to avoid post hoc modifications of the model. Learn how to write effective research papers by employing

E. Practical Evaluation Protocol

Practical evaluation is purposely set apart from the ongoing benchmark demonstration. When constructs and the scoring process are backed up by empirical evidence, a subsequent study can then review whether technology

managers, software practitioners, entrepreneurs and/or other decision makers can comprehend and utilize the CVI results. This should examine the dimension profile to see if these provide an active weakness, if the warnings are comprehensible, if the evidence trail is suitable for review, and if the tool provides information in addition to the technical/business assessment. The collection of any participant and/or organizational data should only take place when the necessary requirements for ethics, consent, confidentiality, access, retention and disposal are met. Negative cases should also be evaluated in the practical evaluation. A strong system should act sensibly in each of the following situations: ↔ When a technically strong aspect is lacking in the organization, ↔ When organization is immature despite the maturity of APIs and ↔ When a high aggregate score hides one weak aspect. Such cases can check if a simple average covers up material risks or if cases should be disallowed, if special circumstances exist, from aggregation. The aim is not to increase the number of assets ready, but to clarify and repeat the conditions for interpreting assets.

IX. IMPLICATIONS FOR RESEARCH AND PRACTICE

A. Research Implications

The framework relates the software engineering and the information-systems views via a shared measure problem. The properties of the software system are called "architecture quality" or "technical debt", and the properties of the organization in which the system will have to evolve are called "organization readiness", "governance" or "organization capability to transform". API maturity is at the middle-ground due to its being both a technical design concern and an ecosystem-participation enabler. Having these dimensions in one model will provide an opportunity to determine empirically if they are distinct and if their combined profile is related to a commercial outcome. Meanwhile, while the main value is to ensure that an integration is not forced aggregation where no evidence or construct is made, the validation gates allow the integration to be made in such circumstances. The study also demonstrates the importance of designing a measurement when using secondary benchmark data in the study. Once numeric variables are provided, a model can be estimated using statistical software, but having an estimate doesn't mean that the variables are measuring the intended concepts. So important, in fact, is the provenance matrix that it is just as critical as the structural model. This is a transferable lesson for research to be applied to public or teaching datasets that have new conceptual frameworks: meaning must be constructed, coding must be done, source context must be identified prior to making an inferential claim.

B. Practical Implications

The proposed CVI is designed to serve as a framework for a pre-scaling discussion for organizations, but is not a substitute for the commercial due diligence process. If a dimension profile can be used it can show the trade-offs. For instance, an asset could be very well-designed both architecturally and API-wise, but poorly-governed or organizationally prepared. This imbalance would be lost if one technical measure were used. On the other hand, a business case might look good and technical debt will allow for risk in the future to be scaled. The framework offers a common format to document the evidence and can be used to inform the prioritization of remediation for the platform prior to embarking on a significant platform investment. The decision-support value is reliant on interpretation which is done in a disciplined way. While the scores can be used to compare and for diagnosis until they are referenced and tested for predictive power, they are not intended as approval or disqualification criteria. The most helpful results might be a score, dimension-level evidence and warnings that alert to areas of uncertainty. This retains the flexibility of management decisions, and provides a more level playing field for management to question the readiness of a prototype to proceed to the next level of platform evolution.

X. THREATS TO VALIDITY AND QUALITY CONTROLS

The framework differentiates four kinds of risk of validity. Construct validity related to the extent to which the chosen indicators are indeed the dimensions of readiness that are hypothesized. The issue of internal interpretation risk arises when there is overlap among predictors whether the effects of each predictor can be estimated separately. External validity is about the ability to generalize results that are gained from the ECSI benchmark to software organizations or other industries. Practical validity refers to the final outputs being comprehensible and applicable in practical situations. The following strengths and weaknesses of the current study should be considered: The design and implementation are best understood, while new empirical and practical evidence is still needed. These limitations are a part of the quality control and not something to be concealed. These risks have a number of controls that follow. Construct definitions are fixed prior to the actual modelling; the provenance is documented at the item level; the direction of the TDM is determined semantically; the transparent comparator is equal weighting; the sensitivity analysis is pre-defined; and the input of valid and invalid items are covered during implementation tests. These controls help to minimize the risk that the framework will be tuned following a favorable outcome. They also facilitate other researchers to reproduce the logic and determine what part of the modelling decision in a change of output. The present preliminary values can therefore only be considered in a limited sense. They demonstrate descriptive and structural diagnostics as possible analysis products of the pipeline and that there is significant overlap in the pattern observed in the benchmark. They do not show a proven causal model, a predictability to success in the market or a universal readiness threshold. The 69.6125 reference score is also a demonstration of the normalizations and aggregation arithmetic for stated assumptions. It should not be used to say whether or not the benchmark case is “very ready for the real market.” The last quality control concept is preservation of evidence. All tables, coefficients/coefficients of variation, scores or figures reported in future submissions must be available from a retained source file, report from a software export or script. If it is not possible to provide a copy of the evidence, the claim should be deleted or clearly identify it as a preliminary claim. This is done to align the publication manuscript with the requirement of the thesis that sets the time of publication based on the maturity of the evidence and not pressure to overstate the lack of evidence.

XI. CONCLUSION

This paper introduces a measurement first framework for evaluating commercial viability of software based digital assets, for progression from prototype to a scalable platform, the Commercial Viability Index (CVI). The framework includes six dimensions of candidate readiness, includes commercial performance as an external outcome, and has clear evidence gates prior to scoring and/or structural interpretation. The preliminary benchmark results indicate that there is a high degree of construct overlap and collinearity and thus the path-derived weights are not considered final. Rather, it has been decided to adhere to an auditable equal-weight 0–100 reference implementation, which explicitly defines failure states. There are 5-unit tests to confirm input-control behavior and implement arithmetic. This contribution is thus a testable and reproducible framework, and empirical construct validation, robustness assessment and practical evaluation are still required before the CVI can be given the status of a validated instrument for commercial-readiness.

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REFERENCES

[1] G. Vial, “Understanding digital transformation: A review and a research agenda,” Journal of Strategic Information Systems, vol. 28, no. 2, pp. 118–144, 2019.

[2] A. Gawer, “Digital platforms and ecosystems: Remarks on the dominant organizational forms of the digital age,” Journal of Management Studies, vol. 58, no. 4, pp. 1108–1114, 2021.

[3] L. Bass, P. Clements, and R. Kazman, Software Architecture in Practice, 4th ed. Boston, MA, USA: Addison-Wesley, 2021.

[4] P. Kruchten, R. L. Nord, and I. Ozkaya, “Technical debt: From metaphor to theory and practice,” IEEE Software, vol. 29, no. 6, pp. 18–21, 2012.

[5] D. J. Teece, “Business models and dynamic capabilities,” Long Range Planning, vol. 51, no. 1, pp. 40–49, 2018.

[6] A. Tiwana, B. Konsynski, and A. A. Bush, “Platform evolution: Coevolution of platform architecture, governance, and environmental dynamics,” Information Systems Research, vol. 21, no. 4, pp. 675–687, 2010.

[7] J. F. Hair, J. J. Risher, M. Sarstedt, and C. M. Ringle, “When to use and how to report the results of PLS-SEM,” European Business Review, vol. 31, no. 1, pp. 2–24, 2019.

[8] J. Henseler, C. M. Ringle, and M. Sarstedt, “A new criterion for assessing discriminant validity in variance-based structural equation modeling,” Journal of the Academy of Marketing Science, vol. 43, no. 1, pp. 115–135, 2015.

[9] OECD/European Union/EC-JRC, Handbook on Constructing Composite Indicators: Methodology and User Guide. Paris, France: OECD Publishing, 2008.

[10] M. Tenenhaus, V. E. Vinzi, Y.-M. Chatelin, and C. Lauro, “PLS path modeling,” Computational Statistics & Data Analysis, vol. 48, no. 1, pp. 159–205, 2005.

[11] S. Nambisan, K. Lyytinen, A. Majchrzak, and M. Song, “Digital innovation management: Reinventing innovation management research in a digital world,” MIS Quarterly, vol. 41, no. 1, pp. 223–238, 2017.

[12] J. Hein et al., “Digital platform ecosystems,” Electronic Markets, vol. 30, no. 1, pp. 87–98, 2020.